Tendai Chimuka

Newark, New Jersey
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SUMMARY

Electrical Engineering student with hands-on experience in circuit design, PCB development, and data analysis for electronic systems. Strong foundation in semiconductor devices, analog/digital circuits, and electronic circuit design, with experience using oscilloscopes, spectrum analyzers, power meters, waveform generators. Interested in electronics and RF

EDUCATION

New Jersey Institute of Technology **Sep 2022 – May 2027**
BS/MS Electrical Engineering - RF + Controls *Newark, NJ*

RELEVANT COURSEWORK

- Analog & Digital Circuits
- Semiconductor Devices
- Signals & Systems
- Electromagnetic Fields
- Electronic Circuit Design
- High Speed Electronics
- Microprocessors Lab
- Energy Conversion

EXPERIENCE

Research Assistant **June 2025 – Present**
NJIT Networking Research Lab *Newark, NJ*

- Co-authored a research paper titled *EcoFlight: Low-Energy Path Finding Through Obstacles for Autonomous Sensing Drones*, selected for publication in the IEEE Xplore Digital Library.
- Supported flight-ready hardware integration by assembling and soldering electrical components, validating power integrity, and conducting functional testing to ensure reliable system performance.
- Designed and built an autonomous drone for data collection missions, using Creo Parametric for component modeling and performed test flights.

Research Assistant **May 2026 – Present**
NJIT Controls Automation & Robotics Lab *Newark, NJ*

- Designed and modified robotic hand components in Creo Parametric for controls and robotics research. Supported mechanical integration of linkage-driven finger mechanisms, actuator mounts, and prototype parts.
- Assisted with robotic hand research by reviewing grasping, manipulation, and control-related papers. Supported early simulation planning for testing finger motion, grasp behavior, and future control strategies.

PROJECTS

NASA Rock-Sat Project | *Altium, Electronic Instruments, C++, Arduino* **April 2026**

- Part of the electrical team for a NASA-selected suborbital research payload that collects electron density data in the ionosphere using a Langmuir probe.
- Collaborated in the design of a 3-stage PCB system utilizing a 4-layer mixed-signal architecture that combines digital devices, precision analog instrumentation, and dual RF telemetry.
- Validated system power integrity through instrumentation testing of 28V-to-5V buck converters and 5V-to-3.3V low-dropout regulators to ensure stable power distribution.

High Power Distribution Board | *KiCAD, Power Electronics, PCB Fabrication* **January 2026**

- Designed and fabricated a high-power Power Distribution Board in KiCad to route a 5–30 V battery input through an XT60 connector, automotive blade fuse, and six independent high-current outputs.
- Integrated an INA226 current/voltage monitor with a 4-terminal Kelvin shunt to accurately measure bus current. Implemented Kelvin sense routing, 4.7 kΩ pull-ups, and decoupling capacitors to reduce noise.
- Built a custom buck-converter power stage around the LM2596 to generate a regulated 5 V rail from a 5–30 V battery input, selecting the switching inductor, Schottky diode, and input/output capacitors for stability.

ESP32 RF Transmitter & Receiver System | *Multisim, Electronic Instruments, C++, ESP32* **October 2025**

- Designed and fabricated a wireless 4-channel RC transceiver system using an ESP32 and nRF24L01 2.4 GHz RF module for joystick and switch-based control data transmission
- Simulated circuits in Multisim, designed and fabricated the PCB in KiCad, and developed Python-based control logic for wireless data transmission and system operation.

TECHNICAL SKILLS

Embedded/Programming: Python, C++, Linux, microcontrollers I/O (GPIO/ADC/PWM), Serial Protocols (UART)
Instrumentation: Oscilloscope, Waveform Generator, Multimeter, Power Supply, Spectrum Analyzer, Logic Analyzer
Simulation/tools: Multisim, MATLAB, LTspice, Simulink, KiCAD, Creo Parametric, AutoCAD, Altium